Chinmay Joshi

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Research Interests

My research interests lie in understanding the **mechanisms of learning** in **deep learning**, particularly in modern neural architectures of **transformers** and **diffusion models**, with a focus on **efficiency** and **robustness**. I am interested in studying these properties as tightly coupled aspects of learned systems, and in understanding how interventions aimed at one can alter the others. Broadly, I am motivated by developing learning methods whose efficiency and robustness are **meaningful in practice**.

Education

Universität des Saarlandes — Saarbrücken, Germany

M.Sc. Computer Science (GPA: 1.7)

- Master's Thesis: **On the Effect of Data Pruning on Memorization**
  (*Advisor: Dr. Rebekka Burkholz – CISPA Helmholtz Center for Information Security*)
  - Studied how **dynamic data pruning** changes **memorization** in deep neural networks, establishing a connection between **training efficiency** and **privacy-relevant model behavior**.
  - Designed and executed **large-scale experiments** on **CIFAR-10** and **ImageNet**, including Feldman et al.'s **memorization estimation** requiring hundreds of model trainings per dataset and **membership inference attacks** for privacy evaluation.

Narsee Monjee Institute of Management Studies — Mumbai, India

B.Tech. in Data Science (GPA: 3.8/4)

Experience

Research Assistant (HiWi) — Dec 2025 – Present

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) — *Kaiserslautern, Germany*

- Developing methods for **subgroup discovery** in **time-dependent panel data**, including **economic datasets**, to identify exceptional temporal segments and characterize them with **interpretable rules**.
- Designed a hybrid approach combining **clustering-based candidate generation**, **KL-divergence scoring**, and **Random Forest-based rule extraction** for detecting and explaining relevant subpopulations.
- Using **SQL** querying to work with large-scale structured temporal data for downstream empirical analysis.

Research Assistant (HiWi) — Mar 2025 – Sep 2025

Fraunhofer IZFP — *Saarbrücken, Germany*

- Analyzed **environmental**, **mobility**, and **geospatial sensor data** to quantify how route choices affect user exposure in urban settings.
- Developed **interpretable route-quality metrics** including UV exposure, sun intensity, and noise burden, and integrated them into **OpenStreetMap**-based routing and visualization components.
- Built an **audio annotation tool** for veterinary recordings that replaced a paper-based workflow, reducing annotation time by up to **80%**.

Research Intern — Apr 2024 – Aug 2024

National Institute of Informatics — *Tokyo, Japan*

- Constructed a **large-scale synthetic dataset** for **3D novel-view synthesis** and **relighting**, generating **150K+ multi-view, multi-light samples** with aligned depth and surface-normal using **Blender**

– Investigated **single-image to 3D reconstruction with relighting** using **diffusion models**; finding that stable relighting with NVS required careful noise scheduling and sufficiently detailed **mesh geometry** to preserve illumination consistency.

Research Assistant (HiWi) — Jun 2023 – Mar 2024
Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) – Project KIMonoS — *Saarbrücken, Germany*

– Built an **algorithmic scheduling framework** for multi-vehicle shuttle systems under dynamic demand.
– Implemented routing and scheduling methods in Python and evaluated their effect on passenger waiting time and seat utilization through extended **SUMO simulations**.

Computer Vision Intern — Jan 2022 – Apr 2022
Aurify Systems Pvt. Ltd. — *Mumbai, India*

– Developed **computer vision pipelines** for retail and warehouse monitoring, including bin detection, shelf-state analysis, and CCTV-based activity recognition.
– Built lightweight solutions using **CNN-based object detection**, **template matching**, and **non-maxima suppression** for resource-constrained deployment.
– Implemented a **pose-based activity recognition pipeline** using **AlphaPose** and SVM, achieving **91% accuracy** in controlled field evaluations.

Selected Projects

- **PCA-LSTM for Algorithmic Trading** (Publication): Proposed a **quantitative trading framework** for Indian large-cap equities using tick-level time-series data, combining **PCA**-based compression of 84 technical indicators, **LSTM**-based signal generation, and **portfolio optimization**. In backtesting on the **NIFTY50**, the method outperformed buy-and-hold by nearly **20 percentage points**, with the resulting portfolio achieving approximately **58% return** at **0.34** volatility. (Code)

- **Hierarchical VAE for Image Compression**: Implemented a **hierarchical variant** of a VAE-based image compression model following Ballé et al., achieving approximately +**5 dB PSNR over JPEG** at similar bitrate through architectural modifications. (Report)

Skills

Programming: Python, SQL, R

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Hugging Face

Tools: Git, Docker, MLflow, Weights & Biases, SLURM, SUMO

Areas: Machine Learning, Deep Learning, Time-Series Modeling, Geospatial Data Analysis, Interpretable ML, Computer Vision (2D & 3D), Pattern and Subgroup Mining, Causality

Languages: English (Native), German – Learning Actively (Listening/Understanding: Intermediate; Writing: Intermediate; Speaking: Beginner)

Volunteering

- **Teaching Volunteer** — The Satsang Foundation
  Taught Physics, Math, and English to students from the tribal community of Aarey Colony, Mumbai.

- **COVID-19 Relief Volunteer** — The Satsang Foundation
  Supported food distribution efforts for communities economically affected during the COVID-19 pandemic.